ISSUES AND RESOLUTION - PRESSURE IN LIQUIDS

This document is a collection of issues that were found as a result of the study of pressure, pressure gauges, laws of pressure, accuracy and precision of pressure and performance metrics of pressure.

Medium: Liquid. All of these issues are studied and explained specifically with respect to liquids. The example of a liquid used to help understand these issues is water.

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Issue 1: Pressure Gauges measure Static Pressure and Gauge Pressure - **Page 02**
A description of why pressure gauges are not a measure of anything and they are not measuring "Static or Gauge Pressure".

Issue 2: Pressure in fluids (liquids) = (Static Pressure, Dynamic Pressure) - **Page 10**
A description of why pressure in liquids can never be "Static & Dynamic Pressure" and there can never be a "total pressure component" that equates to "Static + Dynamic Pressure".

Issue 3: Hydrostatic Pressure = ρgh - **Page 17**
A description of why hydrostatic pressure is not a function of height and what does it actually depend on by explaining flow dynamics.

Issue 4: Absolute Pressure = Gauge Pressure + Atmospheric Pressure - **Page 24**
A description of how "Gauge Pressure" is not relative to "Atmospheric Pressure" and why the relation of "Gauge, Atmospheric, & Absolute Pressure" does not hold true.

Issue 5: Pressure is transmittable, Pascal's Law - **Page 30**
A description of why "Pressure" is not a transmittable quantity which forms the basis of Pascal's Law and how "P" is unnecessary in "P = F/A".

Issue 6: Darcy Weisbach's Equation of Pressure Loss in Pipes - **Page 37**
A description of why Darcy's equation of "Pressure losses" due to friction is first based on the wrong understanding of "Friction" and second not calculating any actual water loss in/ from pipes.

Issue 7: Accuracy and Precision of Pressure Gauges - **Page 44**
A description of why ASME and EN standards of accuracy and precision for "Pressure Gauges" are unnecessary based on what "Gauges" actually measure and why certain design standards for the accuracy of the "Gauges" actually are not driving towards a precise reading of "Static Pressure".

Issue 1: Pressure Gauges measure Static Pressure and Gauge Pressure

Issue:

Pressure Gauges measure:

1. Static Pressure = Pressure due to random molecular motion and collisions of molecules in fluids which translates into thermodynamic pressure exerted onto any surface such as walls of any enclosed container like tanks, pipes, bottle, riverbeds etc. This is measured perpendicular to the surface to get the exact or true thermodynamic pressure. This thermodynamic pressure against surface is without taking into consideration impact of fluid velocity.
2. Gauge Pressure = It is the pressure relative to the atmospheric pressure. How much the pressure of a liquid is at zero point or above/ below the zero point with respect to atmospheric pressure.

Resolution:

The reading that is indicated on every possible gauge, is precisely a measure of:

1. x volume of water as a whole entering the bourdon tube or hitting the face of diaphragm or bellows, causing change in the shape or size of a bourdon tube, face of the diaphragm or the external surface of the bellows translating into a reading or conversion of such elastic deformation into an electrical (capacitance) signal thus transforming into a digital reading.
2. This is not a measure of anything but at best a correlation to the presence or absence of liquid.

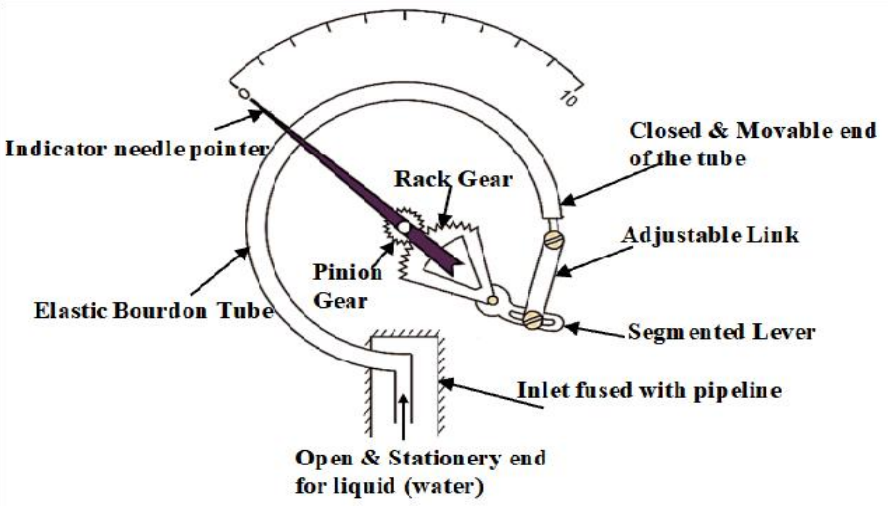
Why:

To understand what a pressure gauge measures and what the reading indicated on the dial means consider the working of a bourdon tube pressure gauge that is most widely used for many pressure vessels. Example used here is an example of a pipeline carrying water.

Mechanical components of a bourdon tube pressure gauge:

1. An inlet that is fused with the pipeline (or any pressure vessel that its connected to).

2. An elastic bourdon tube, which changes its shape and size once it is filled with appropriate amount of water (Or any liquid). It has an open & stationary end connected with the gauge inlet and a closed & movable end.
3. An adjustable link with two ends, whose one end is connected with the closed and moving end of the bourdon tube.
4. A rack gear is connected with the other end of the adjustable link via a segmented lever.
5. A pinion gear that rests on the top of rack gear.
6. An indicator needle pointer that rests within the open space of the pinion gear.



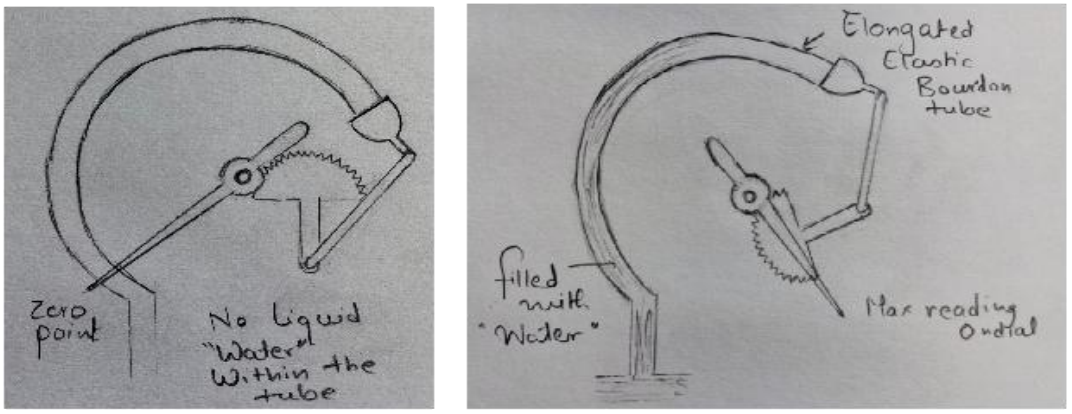
Working of the Pressure Gauge:

1. The section of the pipe is not filled with water yet and the gauge's indicator needle pointer is at 0.
2. Once the pipe gets filled with water there are two ways the water can enter the inlet of the gauge,

   If the water in the pipe fully packs the space, in this case water enters the inlet of the gauge easily.

   If the water in the pipe doesn't fully pack the space, but a spike of volume of water manages to enter the inlet of the pipe.
3. The inlet of the pressure gauge that is fitted on the upper surface acts as an outlet for water in the pipe.
4. As the water in the pipe rushes inside this inlet of the gauge, certain volume of water enters the elastic bourdon tube.

5. Depending on how much the water has entered the tube, the tube changes its shape and size, in this case it straightens i.e. curvature of the tube widens and it's closed & moving end pushes more outward and away from its baseline position.
6. Based on this movement of the closed and movable end of the tube, the adjustable link is pushed more outward from its original position.
7. Due to this outward movement of the adjustable link, the rack gear moves in a direction where its gear teeth are facing the elastic bourdon tube.
8. The rack gear movement causes the tiny pinion gear to move in the clockwise direction as shown in the figure.
9. This clockwise movement of the pinion gear causes the indicator needle pointer to move in clockwise direction.



Interpretation of results based on the working of the bourdon tube gauge:

The volume of water that enters the elastic bourdon tube acts as its own whole body, as the volume of the water that enters the tube is not the whole volume of water that exists within the entire pipe. The water that enters the elastic bourdon tube is a separate water body (As the body that contains the water changes) and the water that exists within the pipe is a separate water body (More on why separate water bodies, refer [Issue 2]). And this whole volume of water within the elastic bourdon results into the change in the shape of the tube.

Thus,

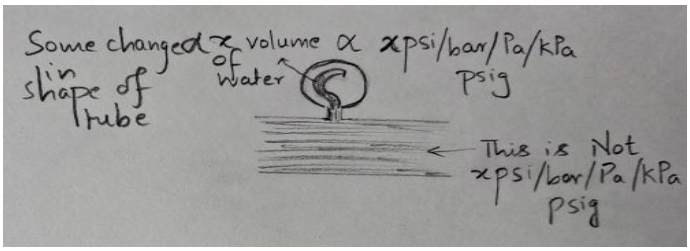
1. Change in the position of the indicator needle pointer = widening of the curvature of the bourdon tube

2. Widening of the curvature of the bourdon tube = x volume of water inside the bourdon tube (as in some weight of water inside the tube)

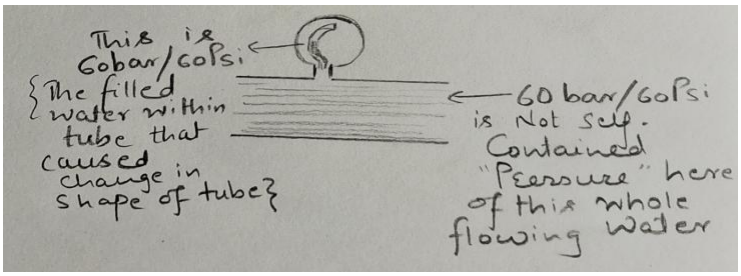
At any instant, if x volume of water rushes inside the tube,

Amt of change in shape/ size of the tube $\propto$ Weight of x volume of water as a whole within the tube

Thus, the reading of x psi or y bar on the dial of the bourdon tube gauge means how much the bourdon tube changes its shape and size proportional to volume of water as a whole body within the bourdon tube.

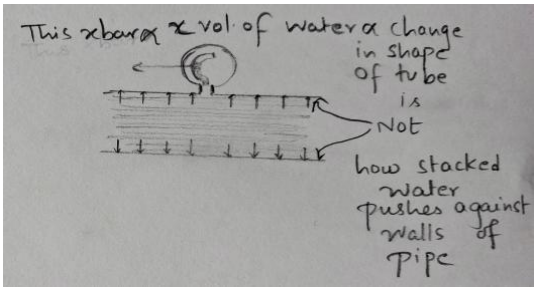


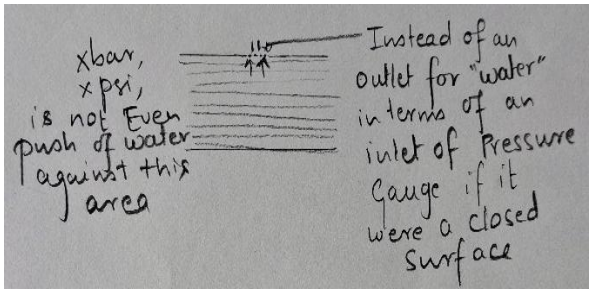
So, in the water pipeline example, a bourdon tube gauge reading 60 bar is not an internal or static or thermodynamic pressure, an inbuilt pressure due to the molecules of the liquid (water) value captured at the instant showing this reading but the value of how much the bourdon tube changes its shape/ size based on the weight of the volume of water as a whole within the tube.



This reading shown on the dial of the gauge is also not:

1. How entire amount of water pushes against walls of the pipe.
2. How amount of water pushes against the surface area of the upper wall of the pipe, if there were a closed surface instead of an inlet for the bourdon tube.





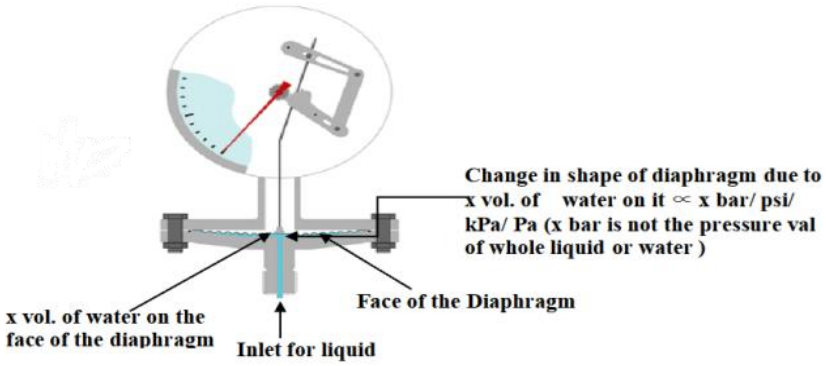
This reading is also not with respect to the surrounding atmospheric pressure because:

1. As seen in the mechanical components sub-section, there is no air vent component that allows the atmospheric air within the internal machinery of the gauge to be compared with “weight of volume of water within the bourdon tube”.
2. As seen in the working of the gauge sub-section, the working of the gauge relies solely on the mechanics of internal machinery upon entrance of certain volume of water inside the tube.
3. And even if hypothetically there were an “air vent” (which is not present in any of the pressure gauges or sensors, refer [Issue 4] for this), the comparison doesn’t make any sense as the bourdon tube reading is the reading of the change in the shape/ size of the tube due to presence of the volume of liquid within the tube and not the “pressure of the liquid” so the difference wouldn’t be measuring anything substantial with respect to the “pressure of the system”.

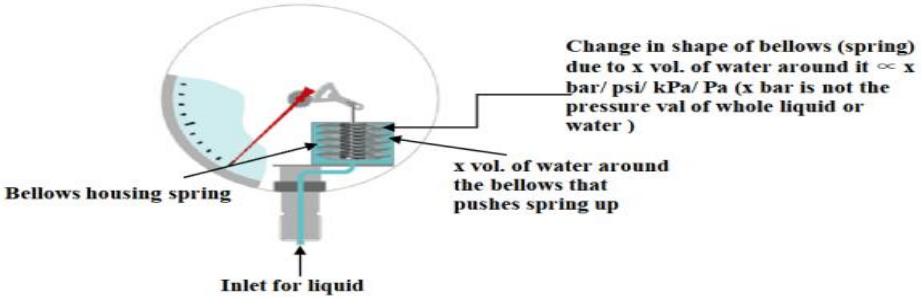
+x psi on the gauge is not a measured value of pressure in the gauge above the 14.69 psi (101.325 kPa) or -x psi on the gauge is not a measured value of pressure in the gauge below 14.69 psi (101.325 kPa). (Detailed description on why the relation of “Absolute Pressure = Gauge Pressure + Atmospheric Pressure” doesn’t hold is stated in [Issue 4]).

This result is same across any pressure gauge, description of the mechanics of different pressure gauges is as follows:

Diaphragm pressure gauge: Here, instead of bourdon tube there is a diaphragm that is attached to internal similar indicator needle pointer machinery. As y volume of water hits the face of the diaphragm, the reading that is shown on the dial is about how the face of the diaphragm changes its shape directly based on the amount of water that collided with the face of the diaphragm and not the “pressure of liquid” as whole.



Bellows pressure gauge: There is an arrangement of elastic metallic cylinder with curves (convolutions) called as bellows with spring within. This bellows changes its shape with the contraction/ relaxation of the spring within; with liquid contained in the case that solely houses the structure of bellows and the spring within. The reading that is shown on the dial is about the volume of water that is within the case that contains bellows elastic structure and how much this volume of water as a whole presses on the external surface of the bellows and not the “pressure of the liquid” as whole.



Vacuum or Negative pressure gauge, Compound Gauges: Vacuum or compound gauges, are usually of type bourdon, diaphragm or bellows. Considering an example bourdon tube vacuum or compound pressure gauge, as the inlet of such a gauge rests on the area that is neither filled with air or water, the air or water within the elastic tube of the gauge will move out in the space of the pressure vessel where there is no fluid medium (air or water) through the inlet of the gauge that is fused onto the surface of the pressure vessel. Due to this the curvature of the elastic bourdon tube reduces than its normal curvature. And this reduction of the curvature of the tube results in the anticlockwise movement of the gear machinery and thus the indicator needle pointer. So the reading shown on the vacuum or the compound gauge is of how much the tube changes its shape in a way the curvature of the tube is shrinks and not

the amount of emptiness or vacuum in the pressure vessel. And this reading too is not with respect to the atmospheric pressure or how much the reading is below the atmospheric pressure as the mechanism of the reading solely relies on the internal mechanics of the respective gauges.

Differential pressure gauge: This can also be of type bourdon, diaphragm, or bellows. There are two inputs, two different inlets to measure the pressure difference. But this "pressure difference" is not difference in the "pressure at two different points" but in case of a bourdon tube type differential pressure gauge, the reading at point A is reading of how much the tube changed its shape due to x volume of water inside the tube. The reading at point B is reading of how much the tube changed its shape due to y volume of water inside the tube. These both readings have practically no relation to each other as gauges at point A and B are not reading the "pressure" of the whole water body within the pressure vessel (such as a pipe) at two distinct points, but the gauges at point A and B are measuring its own separate changes in the elastic material with respect to respective volume of water within each elastic tube at point A and B.

Example:

"70 bar" is read on the delivery/ discharge head of a multistage centrifugal pump, this value is considered as the measurement of the discharge or delivery pressure as in the pressure contained within the whole water at the discharge/ delivery head but the reading actually indicates:

"volume of water inside the tube" that caused the deflection of the tube, as shown in the figure

The idea of measuring any kind of pressure such as discharge/ delivery head pressure, pressure of a medium - high pressure water pipeline or even suction pressure at the pump inlet doesn't make any sense because of how the gauges work and what they really measure. So if a gauge at a suction pressure indicates a value either extremely close to the zero point or below zero point the reading is actually not an indication of anything significant but at best can be correlated with presence or absence of liquid or water in these examples.

Related issues:

[Issue 2] - Pressure in fluids (liquids) = (Static Pressure, Dynamic Pressure).

[Issue 3] - Hydrostatic Pressure = ρgh.

[Issue 4] - Absolute Pressure = Gauge Pressure + Atmospheric Pressure.

Referenced Resources:

"https://tameson.com/pages/bourdon-tube-pressure-gauge"

"https://bcstgroup.com/compound-gauge-measure-pressure-guide"

"Pump Discharge Pressure Gauge Placement"

"https://bcstgroup.com/how-to-use-a-water-pressure-gauge-in-an-industrial-setting"

"https://tameson.com/pages/atmospheric-absolute-gauge-differential-pressure"

"https://www.iqsdirectory.com/articles/sensor/pressure-sensors.html"

"Pressure Gauge Types"

"https://tempsens.com/blog/pressure-gauge-types"

Issue 2: Pressure in fluids (liquids) = (Static Pressure, Dynamic Pressure)

Issue:

1. Static Pressure = Pressure due to random molecular motion and collisions of molecules in fluids which translates into thermodynamic pressure exerted onto any surface such as walls of any enclosed container like tanks, pipes, bottle, riverbeds etc. This is measured perpendicular to the surface to get the exact or true thermodynamic pressure.
2. Dynamic Pressure = Pressure due to speed of the fluid molecules in the direction of flow when the liquid is flowing.This is measured in the direction of the flow due to its dependence on the speed of the liquid.
3. Both are pressures at the molecular level.
4. Total Pressure of a liquid system = Static Pressure + Dynamic Pressure.

Resolution:

1. Weight of a liquid as a whole is always with respect to a surface (surfaces), as the weight can be realized only when its contained.
2. Liquid is not a closed system, even if same water is flowing through different containers, each container with that liquid is its own system i.e. Static Pressure of water in one system ≠ Static Pressure of same water in another system, as it is about weight certain volume of water exerts on the surfaces of container and not “Weight of Water”.
3. There cannot be two components “Static and Dynamic Pressure”, that are a part of the whole or “Total Pressure”.

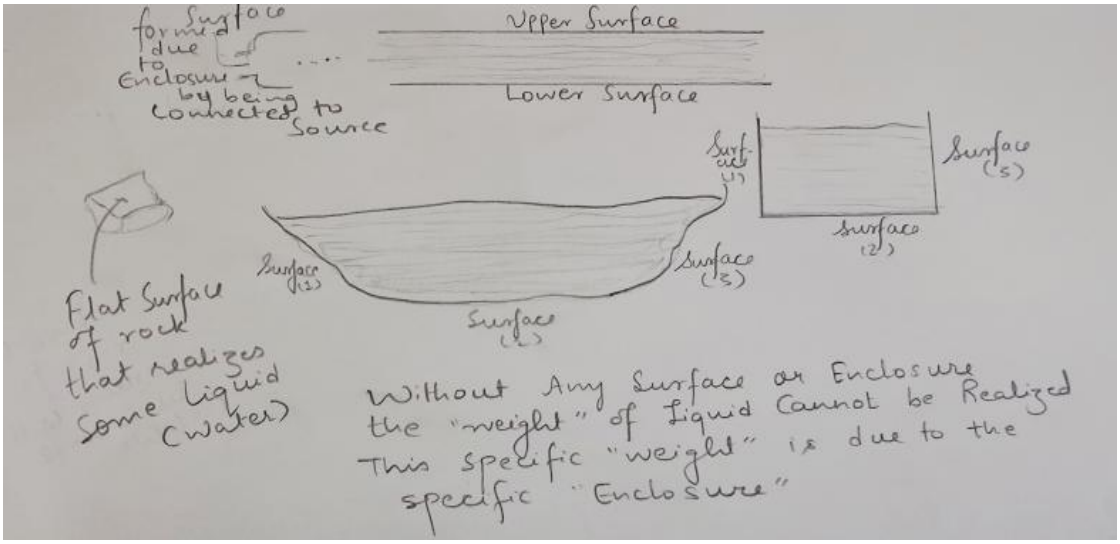
Why:

Dependency of weight of water entirely on the container containing it:

The weight of water exists only when the water is contained with at-least one surface. That is the certain volume of water does not have its own weight unless its contained in a space wide enough or narrow enough to hold this certain amount of volume.

Examples include:

1. Water falling from the top of the the mountain hitting a rock in its way - enclosed by one surface. Here weight of the water is the weight the certain volume of falling water puts on the surface of the rock.
2. Water in an ocean - enclosed by three surfaces - far stretched left and right surface and a bottom surface holding the water contained in it within that space. Here the weight of the water is the weight the volume of water exerts on the far stretched left and right surface and the bottom surface of the ocean.
3. A Water tank - enclosed by three surfaces - The weight of water in a tank is the weight this specific volume of water exerts on all the three surfaces of the tank (left, right, and the bottom).
4. A water pipe - enclosed by three surfaces - Of which upper and lower surface are pipe surfaces and as the pipe stretches from a source and pipe is attached to it, it can act as a third closed surface. The weight of water in a pipe is the weight the volume of water contained across the entire length of the pipe exerting majorly on the upper and lower surfaces of the pipe.



Connecting all of these four examples together by considering the water that falls from the top of the mountains is contained in a ocean, the water from the ocean is contained within a tank, the water from the tank is contained within a pipe. In such a case the water flowing through each of these distinct water bodies is same due to this connection. Now that the water flowing through each of these distinct water bodies is same, the way this same water interacts with these four distinct water bodies is entirely different. And this difference in interaction is due to the difference in the volume of water that each of these four water containers can hold. This

different capacities to hold the water directly leads into different capacities to exert the weight on the surfaces of container containing it. As the “weight of the water” is not an “inbuilt weight of the same water” that flows through all these four water bodies.
To elaborate,

Weight of same water falling on the rocks ≠ Weight of same water within the oceanbed

Weight of same water within the oceanbead ≠ Weight of same water within the tank

Weight of same water within the tank ≠ Weight of same water within the pipe

So if the water in tank is being pushed into a small outlet from the tank, then the water in the tank will be a separate body from the water that exists in the small outlet from the tank. And how the water that acts forward or acts on a surface from the small outlet will be entirely based on:

1. Volume of water in that outlet
2. The cross-section of the outlet (Or the surface the liquid comes in contact with)
3. How fast or slow is the flow of the water (speed)

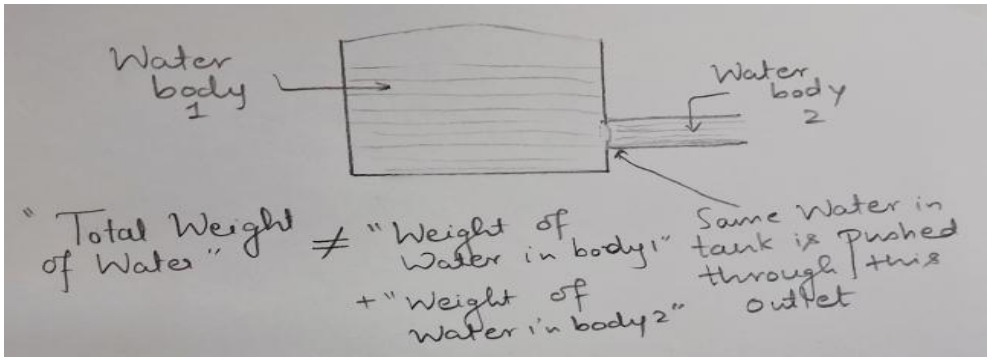
Although if the water flows from the small outlet or not is entirely dependent on the existence of the volume of water within the tank, but when the water exists in the small outlet and how it puts its weight onto the another surface is entirely dependent onto the properties of the small outlet and the volume of water within that outlet and has nothing to do with how water put its weight onto the surfaces of the tank as its contained.

Due to which, water (or any liquid) cannot be considered as a “Closed system” in itself, because a “Closed system” is:

A system in which a total of a certain quantity = one part of a quantity in one condition + another part of a quantity in another condition.

Water in the tank and the same water from the tank into the small outlet is not a “Closed Water system” that can be divided into two parts of whole in a way that:

Total weight of water = Total weight of water in the tank + Total weight of water in the small outlet



There is no component of "Total weight" when the "same water" flows from one water body to another as,

Water in one container containing it = One separate system

Another container containing same water = Another separate system

No "Static or Dynamic Pressure":

Static pressure is taken as a self-contained property of the liquid, where the thermodynamic or static pressure exists due to the random molecular interactions with the surfaces of the container. And this "Static Pressure" of a liquid does not or will not change irrespective of if the same water is now contained within a different container. This completely disregards how any liquid (water in this example) behaves as a "whole". Depending on how liquid (water) behaves as a whole, how containment of the liquid as a "whole" makes it possible for the liquid to exert the weight. And how this "exertion of weight" changes based on the liquid body, there cannot be a room for "Static Pressure" of a whole water body as a part of the whole "Pressure" quantity.

Because, in a water tank - small outlet example, where small outlet is eventually leading to a pipeline,

Water in tank is not under "Static Pressure" aka pressure at rest exerted by liquid molecules contained within the liquid, a part of the "Whole Pressure". The water as a whole (which is not moving) exerts some weight on the surfaces of the tank . And this weight exerted is just that not a part of a whole and not a self contained property.

Now,

The weight the certain volume of water imparts as it rushes out from the small outlet on the upper surface of the pipeline is also just that, the"weight" the certain volume of water imparts as its rushing out from the outlet not a part of the whole. The supposed "Static Component" measured perpendicular to pipeline as the water is pushed from this outlet to the pipeline is its

own separate system (When used any measuring device) and is not same as the supposed "Static Component" in a tank as it's an independent water body.

Same logic applies to having no room for separate "Dynamic Pressure" as another part of the whole. As water flowing out of a pipeline with a certain speed when it comes across a surface onto which this exact weight of volume of water is imparted (with its exact speed), this is a separate property of this volume of water that flowed through a specific cross-section of the pipeline with a certain speed at this instant of time. This is not "Dynamic Part" of the "Whole Pressure", the water exerts in a tank from which the water flows into through a small outlet into a pipeline and the water flowing out of a pipeline.

This leaves no room for "Pressure as a whole" self-contained total quantity of a liquid.
That is,
No Total Pressure = Static Pressure + Dynamic Pressure.

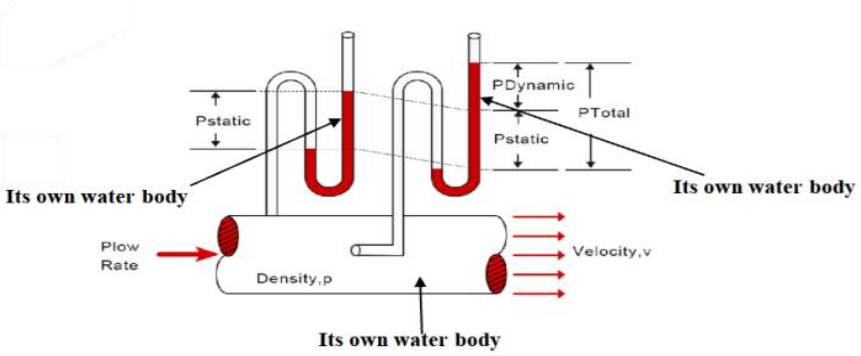
Example:

A pitot tube is often considered as an accurate way of relating "Static" with "Dynamic Pressure" and thus calculating flow velocity from the acquired reading of "Dynamic Pressure" from the "Stagnation or Total Pressure". But it fails largely because,

It tries to measure "Static Pressure" by placing the outlet perpendicular to the flow to capture the "Static Component" of the water.
It measures the "Stagnation or total Pressure" by placing the outlet in direction of flow and making the flow stop at an instant regarding how hard the water exerts the weight on the surface as "Static + Dynamic Component".
Now this "Static Component" is subtracted from the "Total (stagnation) Component" to get the "Dynamic Pressure" and flow velocity of the fluid is calculated from this dynamic pressure.
When there is actually no relation and no importance of what is being measured by the tube,
Static component $\neq$ Self contained pressure as whole acting in all directions
= Random spike of volume of water that entered the measuring tube

Stagnation ≠ Static + dynamic

Component = Sped up spike of volume of water that entered the measuring tube



Subtracting static from this stagnation component is not a measure of anything as each of these "supposed measurements" are its own separate measurements (systems):

1. The "Static component" is an individual behaviour of how certain volume of water as a whole entered the tube.
2. The "Total Component" is an individual behaviour of how certain volume of water as a whole exerts some weight based on its volume, speed and the cross-section through which its flowing.

And addition of these two components does not form a whole "Pressure" component that is flowing through a cross-section of a pipe in case of this pitot tube example.

The same logic actually is applied to gauges "measuring" "static pressure" as seen in [Issue 1], where "60 bar" is deflection of the elastic material due to a volume of spike in water that entered the tube, which again is not a measure of anything but at best an indication of presence or absence of water.

Related issues:

[Issue 1] - Pressure Gauges measure Static Pressure and Gauge Pressure.

Referenced Resources:

"https://www.supmeaauto.com/training/static-pressure-vs-dynamic-pressure-vs-total-pressure"

"Bernoulli's Principle"

"https://cdsentec.com/pitot-tube-flow-meter-how-it-works"

Issue 3: Hydrostatic Pressure = ρgh

Issue:

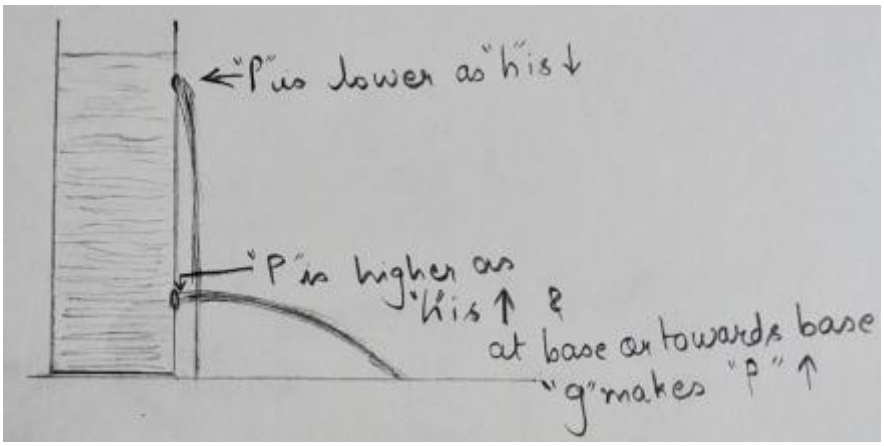
1. Hydrostatic pressure is considered a function of height (depth) of a fluid for pressure at that point.
2. The pressure increases with depth due to gravity.
3. The pressure is same at any point if its at same height irrespective of the size of the container and volume inside the container.

Resolution:

1. The weight at any point within the liquid is not a function of height but a function of volume.
2. The weight at any point on the horizontal surface placed within a liquid at a certain depth does not automatically equate to weight of liquid on a point on the left or right surface of the container in line with the horizontal line.

Why:

Missed Point in the Experiment of the hydrostatic pressure as function(height, gravity):
Consider the most widely used example of a vertical bottle with two holes/ outlets one at point slightly above the bottom surface and one at the top as shown in the following figure:



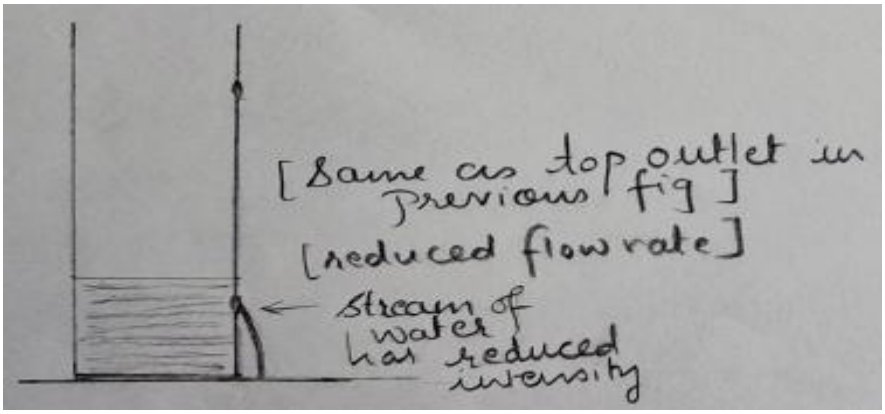
The water is filled till a point just slightly above the top outlet. As the water is allowed to pour out from each of these outlets at the top and at the bottom, the flow of water is higher at the

bottom than at the top. This is the reason hydrostatic pressure is considered as a f(height). That as the depth increases gravity makes the pressure larger at the bottom portion than at the top.

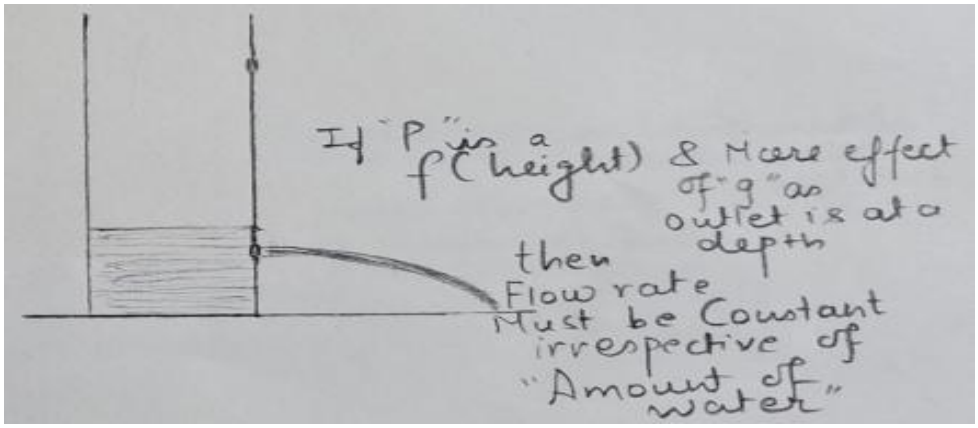
But one largely missed point in this entire experiment is:

That if this pressure at depth is a sole f(height) and gravity is what is making the pressure higher as the movement is towards the bottom surface due to a larger effect of gravitational pull then why,

"The stream of water flow at the bottom outlet reduces drastically as the amount of water or the volume of water above the bottom outlet is very close to this outlet as the remaining water poured out ?"



Ideally, since according to the definition of hydrostatic pressure, the higher pressure at this bottom outlet is solely due to gravity and as a f(height) then water flow rate from this outlet mustn't change irrespective of the level of water above this bottom outlet. So, even if the level of water above this outlet is little, that little amount of water must be pushed out at the same rate that was used to push out the water when its level was close to the top outlet.



Another point is, the drastically reduced water flow at the bottom outlet when level of water above it is very little, is the exact amount of water flow rate that is and will always be observed at the top outlet because the level of water above it is also very little.

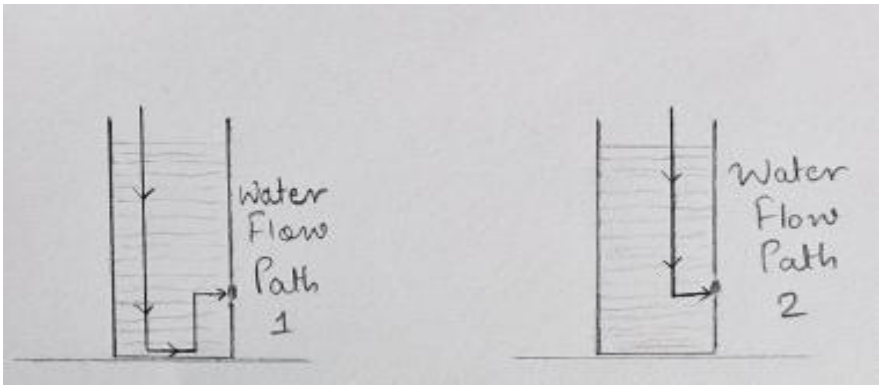
Weight at any point within the liquid as a function of volume:

To explain how it is not about the "higher pressure" at the bottom outlet because the outlet is extremely close to the bottom surface and all about the amount of water above the bottom outlet and how exact same flow rate can be achieved at the top outlet (same as bottom outlet) by having exact same level of water above the top outlet. And how if the cross-section of the bottles (or containers) are different then the weight exerted at any point at a depth within a liquid depends on the volume of water above it and not height. But also how this water pouring out of these outlets is not an indication of pressure but mere flow dynamics.

Consider a container with an outlet slightly above the bottom surface and is filled with water with a certain depth or volume above this outlet as shown in the figure below:

As shown in the figure, there are two paths indicated,

Path 1 of water flow: The one stretching from the level of water at the top -> To the complete bottom surface ->From this complete bottom to the level of the outlet -> From this level of the outlet to the outlet.



Another path from,
Path 2 of water flow: The level of water at the top -> To the level of the outlet -> From this level of the outlet to the outlet.

Now, whenever there is an outlet for water at a depth not just this bottle be it a tank or any other container, the path of water flow from the top level towards the outlet is never the path 1 but always path 2 except for a case when the outlet is already at the complete bottom. The water never flows from top level to the bottom then moves up again towards the outlet that is impossible. The water always flows straight towards the outlet irrespective of the level of water beneath the outlet. The level of water beneath the outlet remains untouched.
And since the level of water beneath the outlet remains untouched,

More volume of water above the outlet = Higher flow rate from the outlet

Less volume of water above the outlet = Lesser flow rate from the outlet

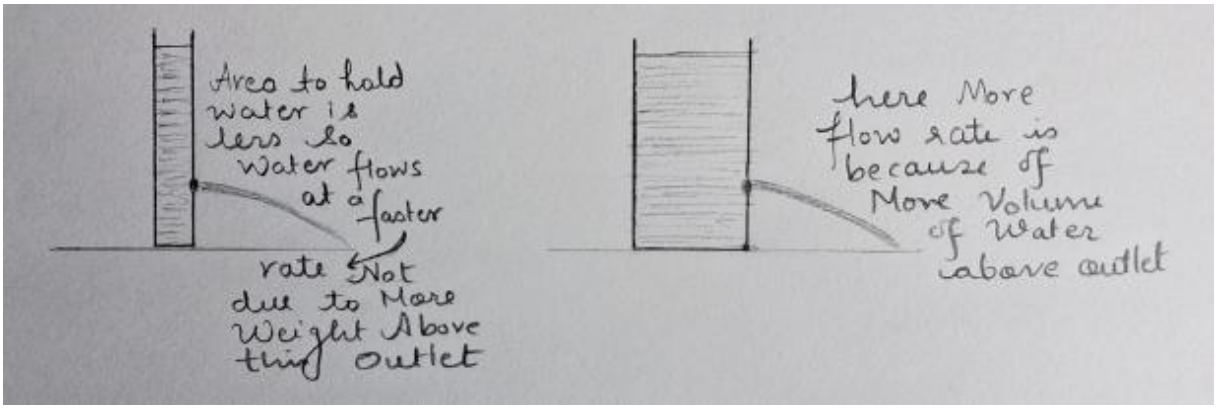
Which shows that, it is not about the pressure at the bottom due to gravity, it is not even about the specific height at which the outlet lays, but it is about the volume of water above the outlet. Gravity holds the water and accumulates on the dominant surface (surface on which the gravitational pull acts), due to which the water flows out from the outlet in downward trajectory. But the flow of water from a specific outlet is directly proportional to the volume of water above it. And the exact flow rate can be achieved at the top outlet by having the exact same amount of water above it as the bottom outlet. Although, bottom outlet will always have an advantage now due to double the level of water above it.

Weight on certain portion at depth x within the liquid $\propto$ Volume of water above the portion

This is simple flow dynamics.

Comparison of weight within the liquid between containers of narrower and larger cross-section:

Now, as shown in the figure below, consider two containers, one with narrow cross-section and one with the cross-section same as the previous example as shown in the figure:



Both of these containers have an outlet at an exact same height from the bottom and from the top. Both these containers are filled with same level of water. Also, the cross-section of the outlet is exactly same in both the containers.

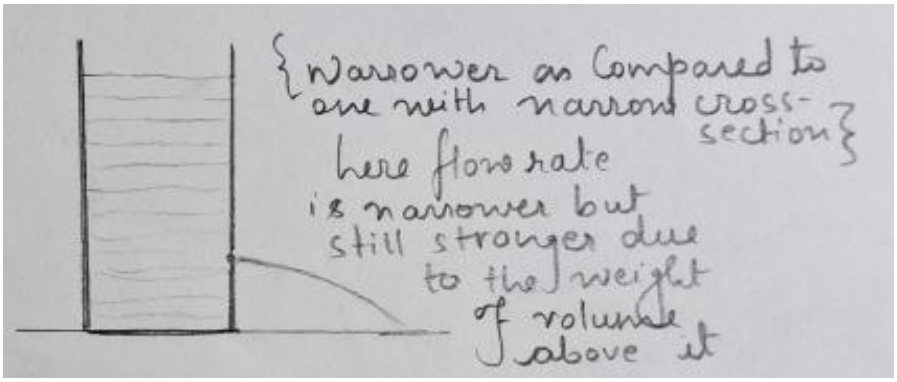
The flow rate from both of these outlets are almost identical, but this is not because both of the containers are filled at exactly same height and have an outlet at exactly same height. This happens because,

In case of a narrow container, the cross-sectional area of the container is small, as in the area to hold the water is less, due to which the volume it contains it also less and since there is an outlet, the water will flow out with a faster flow rate due to inability to hold water for a longer duration because of smaller cross-section. This faster flow rate is not because of the “higher pressure at the point of the outlet” just because it is at the same height as the container with larger cross-section.

In case of another container, the cross-sectional area of the container is larger than the narrower one, as in the area to hold the water is more, the volume it holds is also more, due to which water will stay in this area for a longer duration. Here the stream of the flow rate is stronger not faster. Stronger because of larger volume this wide area occupies right above the outlet and again not because of the height at which this outlet is present.

In both of these cases height has no contributing factor.
Height of one container has no physical relation with the height of another container.

As shown in the figure consider the wider container with narrower cross-section for the outlet while the narrow container has the same cross-section for the outlet.



It was seen that in such case, the wider container has a stronger flow rate but extremely narrower flow stream. While the narrow container has as before a faster flow rate and wider flow stream. This is because of the completely different flow dynamics and not because of the depth each of the outlet is at for each of this container.

As seen from the above examples,

1. The hydrostatic pressure is actually not a function of height
2. There is “weight” on specific portion within the liquid (as seen in [Issue 2] and not self - contained “Pressure”) and is highest at the bottom not due to the pull of gravity, pull of gravity makes water flow downwards, or not due to the height or depth at which this specific portion exists. But the weight imparted on
   any point at a certain depth within a fluid is a function of volume on top of it.

3. Water gushing out of these outlets is not an indication of a pressure (weight) at that specific point, it is the workings of flow dynamics, of how water flows and how fast or slow water moves affecting the flow-rates.
4. But also, the weight of water on a point that is not within the liquid but on the surface of the container, such as these outlets as discussed in the above examples and is at the same depth as any point in line within the fluid, then weight exerted by water on this surface will be entirely different .

Example:

Net Positive Suction Head (NPSH) is the "Suction Pressure Head" calculated at the suction side of the pump to indicate if there is formation of cavitation (When water heats and turns into water vapors and when these water vapors turn back into its liquid form causing a "sudden hit effect with more volume of water").

The calculation of NPSHa makes use of pressure at the suction side, the relation of Absolute Pressure = Gauge Pressure + Atmospheric Pressure (Refer [Issue 4] to understand why this relation will not work); vapor pressure of the liquid, its density and value of "g".

Suction Pressure that is used to calculate NPSH, is the value indicated by the gauge pressure at the suction side. So this "Suction Pressure" value measured at the gauge is not a measure of anything and is neither a "Static Pressure" (Refer [Issue 2] on why there cannot be Static or Dynamic Pressure).

Even when the value of "Gauge Pressure" is not usable, the relation of Abs. Pressure = Gauge Pressure + Atm. Pressure itself is not a correct relation as weight of liquids is always with respect to a surface (surfaces) [Issue 2 & Issue 4].

Value of NPSH cannot be a correct measure of the pressure head value at the suction side of the pump above the liquid vapor pressure.

Related issues:

[Issue 2] - Pressure in fluids (liquids) = (Static Pressure, Dynamic Pressure).

Referenced Resources:

"https://www.varsitytutors.com/practice/lessons/physics/pressure-vs-depth"

"https://byjus.com/physics/hydrostatic-pressure"

"https://www.grc.nasa.gov/www/k-12/WindTunnel/Activities/fluid_pressure.html"

Issue 4: Absolute Pressure = Gauge Pressure + Atmospheric Pressure

Issue:

1. Gauge Pressure indicates pressure relative to the atmospheric pressure.
2. Zero point reading on gauge: Internal pressure = Local Atmospheric pressure.
3. Positive reading on gauge: Positive overpressure, pressure above atmospheric pressure.
4. Negative reading on gauge: Negative overpressure, pressure below atmospheric pressure.

Resolution:

1. Pressure gauges and pressure sensors don't have an air vent to let the atmospheric pressure in for the Gauge Pressure to be compared with the Atmospheric Pressure.
2. Based on how Gauge Pressure is measured and as it is not a measure of anything substantial, it cannot be added with the Atmospheric Pressure leading to the total of the Absolute Pressure of the system.
3. Total weights can only work if they have compounding effect and with respect to a surface and not "Pressure of a system", such as a large riverbed, weight at the bottom of the surface = weight of the volume of water + weight of the volume of air on top of the upper layer of the riverbed (only exists as continued compounded weight and always with respect to a surface).

Why:

Is air vent available in all Pressure Gauges and Sensors?:

The following table lists all the available pressure gauges and even pressure sensors (whose application is strictly not with respect to industrial pipeline networks, pumping systems and water or other liquid systems but listed to cover entire range of available pressure measurement devices), to indicate if they have an opening or air vent for letting atmospheric air in:

Pressure Gauge or Sensor	Internal machinery	Presence of air vent

Bourdon Tube Gauge	External case -> Inlet pipe ->tube (closed & moving end) ->adjustable link ->rack and pinion gear -> needle pointer	There is no opening present to let atmospheric air in
Diaphragm Gauge	External case -> Inlet pipe -> diaphragm and structure to hold diaphragm in place -> Rest similar machinery as above	There is no opening present to let atmospheric air in
Bellows Gauge	External case -> Inlet pipe -> cylindrical bellows with convolutions & spring within -> Rest similar machinery as above	There is no opening present to let atmospheric air in
Capsule Gauge	External case -> Inlet pipe ->pairing of two diaphragms to form a hollow cylinder to hold the liquid -> Rest similar machinery as above	There is no opening present to let atmospheric air in
Compound Gauge	This is either bourdon tube type, diaphragm type, bellows type	There is no opening present to let atmospheric air in
Vaccum Gauge	This is either bourdon tube type, diaphragm type, bellows type	There is no opening present to let atmospheric air in
Manometer	Fluid column with liquid medium -> U tube (main measurement visible via this) -> Source of liquid medium	There is no opening present to let atmospheric air in
Piezoresistive Gauge	External case -> Diaphragm -> resistive circuitry -> reading	There is no opening present to let atmospheric air in
Piezoelectric Sensor	External case -> Diaphragm	There is no opening

	-> electrical charge producing element -> reading	present to let atmospheric air in
Capacitive Sensor	External case -> Main diaphragm -> Capacitive plate -> Rest machinery to convert the signal into reading	There is no opening present to let atmospheric air in
Resonant Wire Sensor	External case -> diaphragm -> tensioned wire and magnetic circuitry -> Rest machinery to translate diaphragm movement into signal into reading	There is no opening present to let atmospheric air in
Optical Sensor	External case -> diaphragm -> light intensity sensing fiber optic circuitry -> Rest machinery to translate diaphragm movement into signal into reading	There is no opening present to let atmospheric air in
Thermal/ Ionization Sensor	External case -> heated filament circuitry to account for changes -> reading	There is no opening present to let atmospheric air in

As seen from the list, none of the most widely used gauges for liquid and water systems have an opening for air for its weight to be compared with the weight of any liquid or water as it enters the tube or hits the face of the diaphragm (even though the gauge measure is a measure of how the specific elastic material changes its shape or size based on the volume of water or liquid that enters the tube or hits the face of the diaphragm and not “Static Pressure”). As in there is no constant and active or even passive comparison of whatever medium of liquid within the internal machinery of each of the pressure gauges/ sensors with respect to the weight of the air surrounding the external casing of the specific measuring device.

Interpretation with respect to Pressure Gauges:

When there is no comparison with the air surrounding the gauges, whatever the gauge measures even though it is not a measure of substantial anything, it is also not referenced with the “Atmospheric Pressure”.
Which means,

When the gauge dial is indicated at zero point, the “pressure” inside the internal bourdon tube or on the diaphragm is not equal to the “atmospheric pressure”. The indicator needle pointer is at a zero point either on a Normal gauge or a compound/ vaccum gauge because the internal machinery is calibrated and designed to be at that position in the absence of external liquid medium. And not because the balance in internal and external pressure balances the indicator needle point to be at 0.

When the gauge dial is indicated at any value above 0, as in a positive reading, the “pressure” inside the internal bourdon tube or on the diaphragm is not greater than the “atmospheric pressure”. The deflection of the tube or the diaphragm or any other elastic material is its own deflection based on the volume of the liquid medium independent of the air pressure. Due to this behavior, there cannot be a positive difference between Atmospheric and Gauge Pressure.

When the gauge dial is indicated at any value below 0 in case of compound or vaccum gauges that are “designed to read negative pressures”, the “pressure” inside the internal bourdon tube or on the diaphragm is not less than the “atmospheric pressure”. And this negative difference is not causing the deflection of the elastic material in an opposite direction from its baseline position. The deflection of an elastic material is its own deflection independent of the air pressure. In this case too, there cannot be a negative “pressure” difference between Atmospheric and Gauge Pressure.

Additionally, as seen from the explanation of what gauges are actually measuring in [Issue 1], and if it even is in the sense of self contained “Static/ Dynamic Pressures” in [Issue 2],

The pressure gauges are not really measuring the “thermodynamic self-contained static pressure”, the gauges are strictly indicating the reading of how much the elastic material changed its shape and size proportional to the weight exerted due to the presence of certain volume of water as a whole in/ on elastic material. Which in itself is not a measure of substantial anything.

So, using this reading and adding it with the atmospheric air pressure equating to the absolute pressure of the entire system, in this case which would be the absolute pressure of a liquid medium within a system such as “water within the pumps or a pipeline network”, does not make sense.

Absolute Pressure of water in a pipe ≠ Gauge pressure of water + Atm air pressure on water

The “Gauge Pressure” of any liquid medium won’t fit in this relationship at all as the “Pressure Gauge” reading is not a measure of anything.

So this equation will never hold whenever there needs to be accounted absolute pressure with respect to gauge and atmospheric pressure.

Also, absolute pressure is defined as absolute pressure of a system when referenced with a vacuum, as in total pressure of a system in itself which also can not be possible. What is only possible is total weight exerted onto a surface (of a body), or on a body.
That is,

1. Not the weight of the liquid body.
2. Not dividing the weight the liquid exerts on the surface (As every containment of a liquid is a separate water body with its separate properties and water is not a closed system [Issue 2]).
3. But the compounded effect of weight with respect to a particular surface, such as a large water body such as a riverbed, weight on the bottom most surface of the riverbed is weight of the volume of water above that surface and continued compounded weight of volume of air on top of the riverbed.

Related issues:

[Issue 1] - Pressure Gauges measure Static Pressure and Gauge Pressure.
[Issue 2] - Pressure in fluids (liquids) = (Static Pressure, Dynamic Pressure).
[Issue 3] - Hydrostatic Pressure = ρgh.

Referenced Resources:

"https://blog.wika.com/en/products/pressure-products/aim-and-purpose-of-a-vent-hole-in-pressure-sensors"

"https://www.iqsdirectory.com/articles/sensor/pressure-sensors.html"

"https://tameson.com/pages/atmospheric-absolute-gauge-differential-pressure"

"https://pressure-sense.com/knowledge/gauge-pressure"

Issue 5: Pressure is transmittable, Pascal's Law

Issue:

1. Pressure as a transmittable, separate property translating into F = P *A
2. "P" is unnecessary in F = P * A

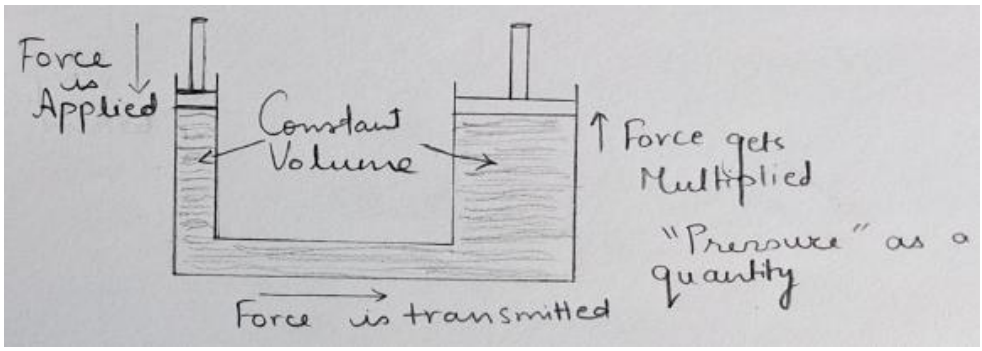
Resolution:

1. In a confined tube filled with liquid (water), where there are no outlets for liquid to push out or drain and each end of the tube is connected to a piston, this whole system acts like a one single unit.
2. Force at the other end of the tube = Force at one end * (Area of the other end/ Area of the first end); as "Pressure" as a self-contained property of liquid is not possible and what is possible is how much weight the liquid as a whole exerts with respect to surface (surfaces).

Why:

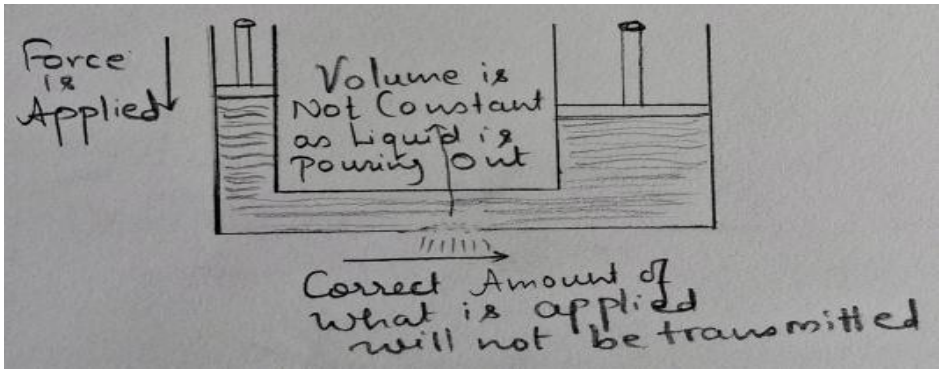
Constant volume confined space making movement of pistons possible:

As shown in the following figure, is a classic inverted tube setup with different cross-sections at each ends and each end connected with a piston:



This is a constant volume confined space because the volume of liquid (water, oil) stays the same without the water pouring out of the space to keep the pistons in its attached positions.

Consider another figure as shown below, where there is small outlet at the middle base portion of the entire tube, the liquid starts pouring out and the ability to push the piston at one end and getting that push transmitted to the other end reduces.



So, again when it is said that "Pressure as a quantity" remains constant in a confined space such as this where the liquid cannot move anywhere but only in a way one or other piston directs, it is actually the "volume" of water that remains constant and confined because the water is not dripping and leaking making the contained confined water body as its separate system and a single unit due to connection of pistons at the both ends. And with respect to the weight this volume of water will exert onto the surfaces of the container would actually be different with respect to each surface (on each surface), i.e. how much weight the liquid (water, oil) exerts on left, right, or the bottom surface with respect to a portion or point on each of these respective surfaces [Refer Issue 3]. Also how much weight the liquid exerts with respect to certain point/ portion within a liquid and this totally depends on the volume of the liquid above it and not height [Refer Issue 3]. And how this weight exerted by the liquid would be different when the the reference portion (i.e. with respect to which the weight gets realized) lies on the left or right surfaces vs the portion/ point lies within the liquid [Refer Issue 3]. "Pressure" as its own separate quantity is not possible and cannot be same and definitely cannot be considered as "Static Pressure" [Can also Refer Issue 2]. It is impossible to have a "Pressure" quantity that gets transmitted and the transmission is an equal transmission.

How "P - Pressure" is unnecessary in P = F/A:

Now, when a Force (F1) is applied to a smaller piston acting on the smaller cross-section (A1) of the tube, the Force (F2) that would be generated on the larger cross-section of the tube (A2) can directly be calculated as,

F2 = F1 * (A2/A1), … equation (1)

A2/A1 in order to get the constant on how many times the cross-section of the second tube differs from the cross-section of the first tube.
And multiplying this constant by the applied force on first piston, F1.

Or to translate into the distance parameter,

If the piston at the second tube needs to be moved in upwards direction by 1 inch, then the piston of smaller cross-section must be moved by (A2/A1) * 1.

Let's say the area of the second tube is 4 times the area of the first then to move the second piston upwards by 1 inch, the first piston must be moved down by 4*1 = 4 inches, or if second one needs to be moved up by 2 inches, then the first one must be moved down by 4*2 = 8 inches.

These relations would work perfectly if needed to use either distance as a parameter or force as a parameter to drive the second or larger piston in the examples seen above or any hydraulic system.

But when a third quantity such as "Pressure" is introduced within these calculations such as,

P = F/A,

Now this equation already translates for a confined space filled with liquid as,

F1/A1 = F2/A2,
Pascal's Law (P1 = P2, Pressure is transmittable)

Or to rearrange this equation to calculate F2, force generated at the other end of the piston, this relation is same as seen in the equation 1.

F2 = F1 * (A2/A1),
This obviously will give the same result as the equation (1) because there is a direct relation between both these forces and cross-sections even without introducing “P”, and relating these forces and cross-sections via a relation of “P1 = P2”.

Now to say that a hydraulic press system operates at 1000 psi,

If this 1000 psi is calculated from only F1/A1 as P = F/A then it is simply replacing F1/A1 in calculation of F2 by another constant value.

Or, if, this 1000 psi is calculated using a gauge or other measurement device then, this value is actually not showing the pressure of the entire liquid (water, oil etc) it is just showing the value of deflection of elastic material due to presence of some volume of liquid a whole as stated in [Issue 1] and this pressure computed from gauge is also not “Static Pressure” and there is no “Static Pressure” as a self-contained property as stated in [Issue 2]. So a value from a gauge cannot be used to calculate the Force at the other end of the hydraulic system (or the tube example in this explanation).

So, just replacing P by F/A doesn’t make sense as what is being applied to the first piston is actually a Force of some amount or even push down in terms of distance and not a pressure of some different value. What this translates into is,

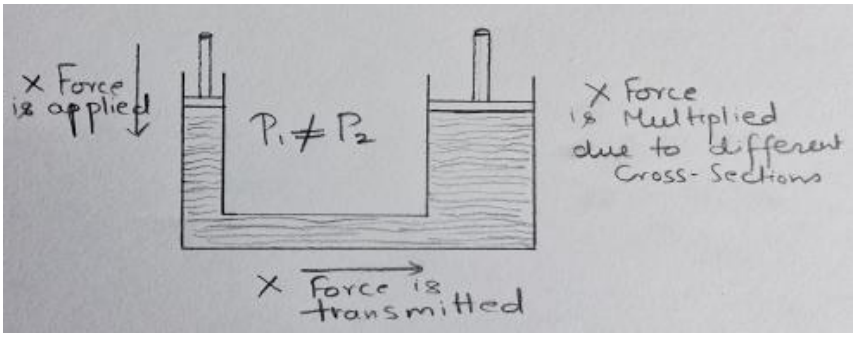
> There can be no separation between the amount of force applied and amount of pressure applied.

As P works only and if only F1 and A1 are known and only in this case it can translate into the almost correct value of F2 generated at the other end of the piston tube example (Or any hydraulic system). And,

“P” doesn’t work if F1 and A1 are unknown, as “P” is actually not an actual separate Pressure that is different from force.

What is transmitted is only force and that force gets transmitted because the volume of liquid stays constant by not leaking or dripping making it a continuous system in itself. It is not "x force is applied" due to which "y pressure gets transmitted" there is no room for this "y pressure" in again what is simple flow and liquid dynamics.

"x force is applied" = "x force is transmitted" (Given volume of water is constant)
"y force is applied" = "y force is transmitted" (Given volume of water is constant)



And not,
P1 = P2, a separate quantity of pressure other than force is not transmitted. As how much the weight certain volume of liquid exerts

This transmitted "force" gets multiplied by the constant factor calculated from the ratio of (area of second cross-section housing a larger piston)/ (area of first cross-section housing a smaller piston)

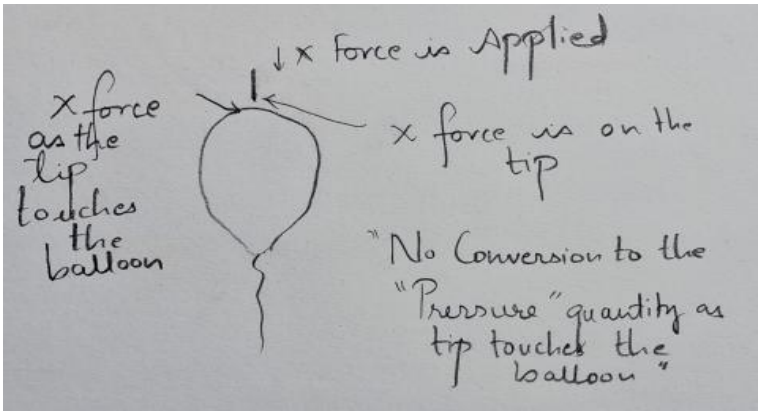
"P = F/A" also not working outside of Pascal's Law:
P = F/A doesn't work within pascal's law and is unnecessary as P acts as an replacement for F1 and A1 if they are known, P = F/A doesn't work outside of Pascal's Law too. Even though its used even for solid objects or to derive hydrostatic pressure (which itself doesn't work as a function of height [Issue 3]), or for water flow in pipes in narrow cross-section, water flows faster due to the space to hold the water is less (a version of this is explained for vertical narrow cross-section [Issue 3]), but by this formula as cross-section decreases the Pressure increases which is again incorrect, because it is talking about "Static Pressure" but according

to weight and flow behavior of liquid (water), and even the parameter of friction it is not the "Static Pressure" that decreases, refer [Issue 6].

For solid objects,
A most widely used example is,
A pin used to blow the balloon and the balloon pops because the force applied translates into high pressure. Aka two different quantities are in interplay in the situation.

But the a small force is applied to the pin let's say x Newtons, now as the tip of the pin touches the balloon the force is still x Newtons at the tip there is no translation into a "high pressure zone" from a tiny applied force at the tip because the area of the tip is small that it is equivalent of a point. The balloon pops because the tip of the pin is pointed. This is a one of scenario that becomes also applicable to any sharp objects. They cut because the edge is sharp not because the area is small if that were the case then the not-sharp area of the cutter would also cut because the area is small so "Pressure" must be larger despite a small force. It doesn't work this way.

Apply "x force" = "x force" is transmitted"
Apply "y force" = "y force" is transmitted"



If applied P = F/A logic to solid objects, for objects that are not sharp or pointy, the force needed to push or penetrate one surface onto other surface, if there needs to be more push or penetration more force must be applied, this exact amount of force is transmitted. No involvement of an extra "Pressure Quantity".

Related Issues:

[Issue 2] - Pressure in fluids (liquids) = (Static Pressure, Dynamic Pressure).

[Issue 3] - Hydrostatic Pressure = ρgh.

Referenced Resources:

"https://byjus.com/physics/pascals-law-and-its-application"

"https://byjus.com/physics/pressure"

"https://www.varsitytutors.com/practice/lessons/physics/pressure-vs-depth"

Issue 6: Darcy Weisbach's Equation of Pressure Loss in Pipes

Issue:

Pressure loss in pipes due to friction between water and pipe walls is:

1. More when the cross-section of a pipe is small and velocity of the water is higher, as more friction.
2. Less when the cross-section of a pipe is large and velocity of the water is lesser, as less friction.

Resolution:

1. Friction is more when the water is fully packed and staying more against the walls of the pipe i.e. larger cross-section and less velocity
2. Friction is less when the water rushes out more in the smaller cross-section of the pipe as the area to hold the water is less
3. Water in pipes does face resistance but these are the actual physical blockages such as x volume of water getting stuck in portion of valve or protruded ridges or something similar or when the flow is against the gravitational pull.
4. Darcy's Weisbach equation doesn't actually calculate the actual frictional losses in the form of actual volume of water lost due to the blockages or the certain infinitesimal amount of water laying on the walls of the pipe (such as those that is seen on the inside of the water bottle once the bottle is empty); it tries to calculate the "pressure drop" due to the reduction in the cross-sectional area which is first not an "actual frictional loss" and second what is called as "pressure drop or loss" is again the effect of flow dynamics of water.

Why:

To understand how darcy's pressure loss due to friction as a f(V, L, D) fails, it is necessary to understand what friction really is.

What is Friction?:

In case of the pipes, friction is said to be increased when pipe gets smaller because the water is being pushed through the narrow cross-section, the friction between the walls of the pipe

and water increases causing “pressure” to drop. But this understanding of friction doesn’t work due to what friction actually is.

Friction is:

How tightly one surface holds onto another surface as in how tightly one surface grips onto another surface.

More grip between two surfaces = More Friction
Less grip between two surfaces = Less Friction

An easiest example of this is the tyres of any vehicle, they are made of rubber and have slight ridges onto them which makes the vehicle grip more onto the surface of the road via tyres. If the tyres were made of really smooth material as smooth as the layer of liquid such as “oil” then the vehicles would be all over the place.

Liquids and specifically water has a thin smooth surface which can glide across or against any surface easily due to its low viscosity, i.e. water has naturally very low friction against the solid surfaces that water rests on. And even though heavy oils which are also liquids, have high viscosity as compared to water, they are still used as lubricants to reduce the “friction” between two mechanical gears so that they run smoothly without getting stuck often.

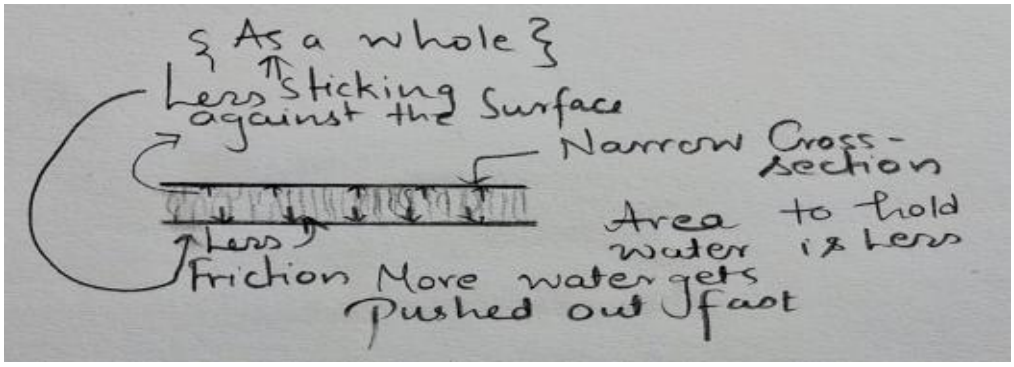
Friction between water and walls of the pipe:
To translate this understanding of friction into what friction is between water and walls of the pipe, it can be understood based on how water flows through different cross-sections of the pipe:

In terms of loss physical volume of water,
In a pipe water is constantly flowing, this flow of water is not restricted (flows with minimal resistance with respect to the surfaces of the pipe in both the cases, when water is flowing in downwards direction due to gravity and water flowing in upward direction due to enough push required to overcome the resistance of gravitational pull) unless it’s a physical blockage like certain volume of water getting stuck in some protruded areas like sometimes the ends and bends of the pipe or getting stuck in valves.

Due to low resistance in the flow of water through pipes unless its physical blockages,
As the cross-section of pipe is smaller, the area to hold the water is smaller, so the water will be pushed out fast from this smaller cross-section. This behavior is independent of if there is a bigger cross-section prior to this smaller pipe area as in independent of the bigger push leading to the faster flow of water from narrow cross-section of the pipe (Consider a narrow vertical water bottle with an outlet close to the bottom surface, let the water flow through this outlet it will be seen the speed of the water flow through this narrow bottle is much higher as the space to hold water is less, water is not going to stay in but push out faster, this behavior is also explained in [Issue 3] to explain the issue of hydrostatic pressure).This leads to following behaviors in a pipe with narrow cross-section:

1. As the water is pushed out fast, the weight the volume of water will put on the upper and lower surfaces of the pipe will actually be lesser.
2. Due to this less weight of water on the upper and lower surfaces of the water, the contact between the water and the surfaces of the pipe will be less.
3. And less contact with the upper and lower surfaces as in even less friction than normal.

Smaller cross-section of the pipe = Less Friction on the upper and lower surfaces of the pipe

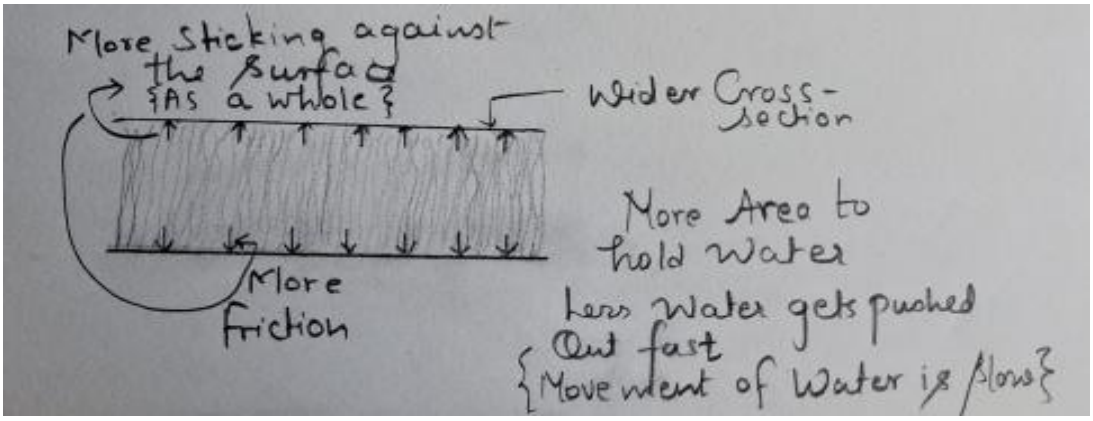


On the other hand, as cross-section of the pipe is bigger, the area to hold the water is more due to which the water rushes out slowly. Since the area to hold the water is more and the water rushes out slowly, this leads to following behaviors in a pipe with wide cross-section:

1. The weight the volume of water will put on the upper and lower surfaces of the pipe will be more as the water remains stacked in this larger area for a longer duration.
2. Due to this more weight of water on the upper and lower surfaces of the water, the contact between the water and the surfaces of the pipe will be more.

3. And more contact with the upper and lower surfaces as in more friction.

Wider cross-section of the pipe = More Friction on the upper and lower surfaces of the pipe



Why Darcy's Frictional Pressure loss is not calculating actual losses based on the understanding of friction:

According to the Darcy Weisbach's equation for pressure loss/ pressure drop which is :

$$\Delta P = f * (L/D) * (\rho v^2/2)$$

The pressure drop or pressure loss is inversely proportional to the internal diameter of the pipe as in the cross-section that would be directly in contact with the water. Meaning for narrow cross-sections the pressure loss is higher.

And, directly proportional to the square of the mean water velocity as in more pressure drop/ pressure loss as the velocity of the water flowing through a certain cross-section increases.

And f, here is the darcy friction factor, that is calculated based on another two parameters depending on the type of flow:

For Laminar flow : It is purely based on Reynold's number (because the flow is considered to be smooth and unaffected by the internal roughness of the pipe) and is calculated as, 64/ Re (Reynold's number)

For turbulent flow: It is calculated based on internal roughness parameter and Reynold's number (using colebrook white equation or moody chart)

Now for laminar as in smooth flows the f value is generally 0.03 or larger and for turbulent as in rough flows the f value is generally 0.03 or lower due to low Re in the former large Re in the latter.

So, a common example of estimating "pressure drop/ loss" would be:

A longer pipe (L is larger) with smaller internal diameter of the pipe (D is small), higher fluid flow velocity (v is larger, v^2 doubles this effect), and smaller friction factor (for turbulent flow) resulting into a "high pressure drop/ loss".

What this "High Pressure Drop or Loss" result means:

1. This "higher pressure drop or loss" is not the actual loss of volume of water when the water comes in contact with the surfaces of the pipe.
2. This "higher pressure drop or loss" is a version of Bernoulli's principle where in, Bernoulli's tries to explain why low pressure (Static Pressure) due to narrow cross-section.
3. This "higher pressure drop" is not because of an increased friction due to smaller internal diameter of the pipe, the friction is actually very small because of the way what friction is. Which is more contact between water and the pipe walls which doesn't happen when the water is flowing through narrow cross-section as water holding area is smaller so more water is being pushed out and due to this less friction.

The result that is estimated from the Darcy's equation of frictional pressure losses in terms of high or low pressure loss is just trying to at best "model" the concept of "Pressure drop" when the cross-section gets narrower and when the velocity of the liquid through this cross-section increases and giving another explanation for this behavior of "Pressure" which is "More or Less Friction" instead of "Static or Dynamic Pressure" which gets used to explain this same behavior of "Pressure" as an interplay of "Kinetic and Potential Energy" by Bernoulli's. Despite of this, the understanding of the friction itself for the Darcy's equation is incorrect to lay any foundation on that. As it is not capturing the actual loosening/ hardening effect of grip of the water with the pipe walls.

Thus, the result of Darcy Weisbach equation practically would be of no use because it is not giving a practical interpretation of the actual water losses when faced resistance/ friction at portion of valves, tees, elbows, entrances and other fittings of the pipe that may act as physical resistance or a form of water blockage that is making the physical loss of water in certain amounts.

Instead of this equation that is used to estimate the "major losses" the most useful and only useful estimations would be in the form that records physical water losses at physical blockages as stated in the last paragraph (valves, tees, protruded ridges etc).

Extras:

The concept of laminar and turbulent flow itself doesn't make sense based on the practical behavior of the liquid (water in this case):

Laminar - When the water is stacked in smooth layers, the first layer of water sticks and fills the smaller gaps on the surfaces of the pipe making the first layer remain stagnant and making rest of the water flow. Laminar flow is said to have less friction because the first layer protects rest of the water from interacting with the roughness of the pipe.

Turbulent - When the water is not stacked in smooth layers, it violently hits against the walls of the pipe and interacts with the roughness of the pipe. Due to this turbulent flow is said to have more friction due to violently interacting with the walls and roughness of the pipe.

According to the behavior of the water there can be only two cases:

1. The water is fully stacked or packed against the upper and lower surfaces of the pipe (More friction because the whole body of water sticks against the upper and lower surfaces of the pipe) (More interaction with the pipe roughness if any).
2. The water is flowing with a high velocity (Less friction because the whole body of water is moving with more velocity the sticking against effect is less) (Less interaction with pipe roughness if any).

In both these cases, water acts as a whole body rather than first layer of water being separate from the rest of the layers or one layer of the water separate from rest of the layers.

Related Issues:

[Issue 2] - Pressure in fluids (liquids) = (Static Pressure, Dynamic Pressure).

[Issue 3] - Hydrostatic Pressure = ρgh.

Referenced Resources:

"https://fm-nitk.vlabs.ac.in/exp/friction-in-pipes/theory.html"

"https://fluidflowinfo.com/darcy-weisbach-equation"

Issue 7: Accuracy and Precision of Pressure Gauges

Issue:

1. Accuracy = How much the value shown on dial is close to the “true pressure value”; allowable error (range of tolerance over a full-scale range).
2. Precision = How often the same measurement can be repeated with the given accuracy. Usually in terms of Linearity, Hysteresis, Repeatability.
3. ASME B40.1 and EN 837-1(2) accuracy grade with classes (+- 0.1% - +- 4 or 5%) for mechanical gauges.
4. Selecting a Pressure Gauge to have a correct pressure range to get accurate readings.
5. Selecting a Pressure Gauge to have a correct unit (psi, bar or both with a dominant/ primary unit).
6. Design Standards for Pressure Gauge Mounting - ASME B40.1 standard for flush (surface) or stem (vertical) on how to fuse the inlet of gauge with the pressure vessel and the pressure gauge should be mounted upright (perpendicularly) to get the correct static reading and to avoid the effects of fluid velocity.

Resolution:

1. Accuracy and Precision does not mean anything, accuracy class of +-0.1% = +-0.5% = … +-5%.
2. There is no known and measured “True Pressure Value” of any water system, and it cannot be back-tracked from the “measured” dial reading. That if gauge is reading “60 psi” and the accuracy class of +-1% is used then the “True Pressure Value” is/ must be 59 - 61 psi, this “True Pressure Value” is limited to mathematical reproducibility and not an “actual range” of an “actual” pressure value.
3. Precision in terms of linearity, hysteresis, and repeatability is strictly with respect to the mechanical hardware working correctly and mechanical wear and tear; has nothing to do with the “reading” in the sense even if the precision is perfect still reading is not even close to “accurate”.
4. The pressure gauge range matters only if the indicator needle pointer moves beyond or below the maximum or minimum value.
5. Choosing a pressure scale with correct unit is not a contribution towards an “accurate reading” as what is being measured is not transmittable to bar, psi, psig, kPa.

6. Designing a gauge inlet so that it is fused (does not come off) with the pipe or other such pressure system in a way its vertical so as to avoid flow velocity effects does not result into an “accurate” result.

Why:

Choosing a correct pressure range = accurate reading?:

As seen in the [Issue 1] that, all pressure gauges measure the deflection of the elastic material that is proportional to just the volume of water as a whole either within or on the material. Then choosing a pressure gauge with any range be it 0 - 100 psi or 0 - 100 bar or 0 - 100,000 psi doesn’t really make a difference because as shown in the figure below are two gauges one with scale 0 to 100 psi and another with scale 0 to 100,000 psi:

In each of these gauges that are shown in the figure, whenever the indicator needle pointer is at the maximum value i.e. at 100 psi and 100,000 psi, both are actually indicating the same thing which is maximum change in the shape of the bourdon tube because of the maximum amount of water within the tube (In case of a bourdon tube pressure gauge) (Same applies to all the remaining gauges either analog, digital or a pressure sensor).

This maximum amount of water within the tube is because of one of the following scenarios:

1. Either the pressure vessel such as a pipeline is fully filled the liquid (water).
2. Either the pressure vessel such as a pipeline is not fully filled with the liquid (water).

In both these cases, it is still certain volume of water within or on the elastic material of the gauge. Where this entry of this volume of water is a bit easier when the liquid is fully stacked vs when the liquid is not, in the case where liquid is not fully stacked a maximum entry of volume of water can still occur if the speed of the water flow is higher and happens to spike through the inlet of the the pressure gauge.

To deduce, the scale of 0 - 100 psi or the scale of 0 - 100,000 psi at best is indicating the presence of water in the pressure vessel with the values of 100 or 100,000 indicating that the liquid (water) is fully stacked or even not fully stacked when the “x volume of water” happens to enter the elastic material or come in contact with it by chance. Thus,

The value of 100 psi or 100,000psi is not showing two distinct quantities of pressure but the same thing:

1. Presence or absence of the water or liquid as the base and best correlation.
2. Correlated indication if the level of water (liquid) is highest when the indicator needle pointer is indicated at the maximum value on the dial.

When the indicator needle pointer moves beyond the maximum value this is not in terms of:

The actual pressure value of the water (liquid) as a whole is beyond that value. In case of a gauge with a range of 0 - 100 psi, indicator pointer moving beyond 100 psi is not because liquid pressure as a whole is > 100 psi, but the pressure vessel such as a pipeline actually has more capacity to hold the water as in wider cross-section and when its fully filled or stacked more volume of water gets acted upon the elastic material of the gauge (in case this happens repeatedly) or one of scenario like a water hammer (where the volume of water happens to reach the elastic material even the pressure vessel such as a pipeline is not fully filled with water).

So selection of a pressure gauge range is valid not in terms of reading an accurate reading but in terms of indicator needle pointer moving beyond maximum value or below the minimum value. Although, selection of a wider or narrower range for the gauge would also not be a measure of an accurate substantial reading based on how gauges work [Issue 1].

Choosing a gauge with correct pressure scale:

Selection of a gauge with correct pressure scale will be of no use practically, as many gauges have dual scales with a primary scale and its respective conversion into the second scale. And what the gauges measure, which is change in the shape or size of the elastic material within any gauge proportional to volume of water within or on the material. This “change in shape of the material” due to “volume of water or liquid” is not transmittable to any of the units of Pressure such as psi, bar, psig, Pa, kPa.

A True Pressure Value:

Accuracy classes and accuracy grades are designed for the Pressure Gauges in order to have an allowable deviation from the true value of the measurement.

Error value = True or Actual Pressure Value - Measured value from the Gauge

But as seen from what the gauges measure, it is not the measurement of a “Static Pressure” value that a gauge is generally designed to measure, [Issue 1] & [Issue 2]. So, it cannot be subtracted from the “True Pressure Value”.

Now, in terms of knowing this true pressure value , which is also a “Component of Static Pressure” (with some deviation from the measured value) as a whole self-contained closed system property within any pressure vessel; this value of liquid is not possible (As this again refers to “Static/ Dynamic Pressure”)as a value of its own, as there is only an estimation of how certain volume of liquid as a whole (As its own separate system) exerts a weight onto a surface [Issue 2].

And this would practically translate into how much stress or weight a pressure vessel can sustain in terms of:

1. Burst Pressure - Which involves physically testing the limits of any pressure vessel to observe for when the pressure vessel begins to leak or burst.
2. Maximum Allowable Working Pressure (MAWP) - This is in terms of how much stress or weight the weakest part of a pressure vessel can sustain. But this pressure too involves estimation of these values in a way its function(height) the way hydrostatic pressure functions. So this may not result into an exact estimation of how much stress a pressure vessel can handle. (Although this is only for Pressure Vessels and not piping design as stated by ASME B31.3 for process piping design standards would take into consideration only design pressure and not MAWP).
3. Operating Pressure - This is in terms of “Liquid Pressure”, aka how much pressure the liquid within any pressure vessel such as water within a pipe has. And the selection of gauges is dictated by selecting the maximum limit of the gauge as (1.5) or (2) * Operating Pressure or to use the rule of middle third i.e. the operating pressure must lie at the 50 or 75% value on the dial of the selected range pressure gauge (This is stated in EN 837-2 but is also used in ASME design standards). This operating pressure value of a liquid will not be of much use this value is with respect to the value as a whole of the entire liquid which is not possible.
4. Design Pressure - According to the ASME Section VIII (UG 21), design pressure is defined as the pressure difference between the inside and the outside of the pressure vessel. That is this must be considered in relation to the absolute and atmospheric pressure. Or the design pressure is some percentage of the operating pressure based on the type of pressure vessel that is being designed.

The most useful value in the practical sense to understand how much load a pressure vessel can withstand would be the value of "Burst Pressure".

The remaining three values of Pressure with respect to how much physical load a pressure system can sustain, may work if a "specific value" is known to be working for "specific system" under certain design conditions. As the actual value of "Operating Pressure" is not useful due to how liquid behaves and exerts its weight [Issue 2]

The value of Maximum Allowable Working Pressure, is a direct calculation from the hydrostatic pressure equation, now this may happen to work as an approximation in certain conditions as certain volume of water can be translated as a height parameter, but the calculation of static or dynamic heads and thus MAWP still needs to account for an actual weight of volume of water instead of height as merely height of one container has no physical relation to the height of another container [Issue 3].

Accuracy Grades and Accuracy Classes for the Pressure Gauges:

Since there is no usefulness of calculating an error from an "Actual Value" of the pressure, the design and usage of the accuracy grades and classes, so that for a high accuracy application a pressure gauge offering minimal deviation from the error, as in a gauge with the highest accuracy class must be selected. And an application where there is no strict requirement for a high accuracy reading a gauge with a low accuracy class can be selected.

But these selections of different accuracy classes would be of no use, as,

> +-0.1% = +-0.5% = +-1% = +-2% = +-4% = +- 5% accuracy class, because neither 0.1% class is of high accuracy and 5% one with least accuracy.

So, +-0.1% accuracy class is not giving a highly accurate reading and a +-5% class is not giving a reading with low accuracy.

There are eight grades designed by ASME standard B40.100 from Grade 4A (+-0.1%) to Grade D (+-5%). There are 7 accuracy classes given by European standards EN 837-1 (For Bourdon Tube Gauges) from +-0.1% - +-4% (high to lowest accuracy). Whereas EN 837-3

states accuracy standards for Diaphragm and Capsule gauges from accuracy class +-0.6% - +-4%.

And according to the interpretation of why +-0.1% class is not a class of highest accuracy and +-5% not a class of lowest accuracy, these definition of different grades and standards and selection of the gauges based on these standards for specific application won't add to the accuracy of the reading.

Precision of the Pressure Gauges:

As the accuracy grades and classes make no difference to the accuracy of the reading from the gauges, the precision of the gauge can only be considered with respect to its correct mechanical functionality. Which would be in the following form:

Linearity: It is stated as linear proportionality to deflection of the elastic material. This would be considered only in terms of deflection of the elastic material such a bourdon tube, diaphragm or bellows.

Hysteresis: It is stated as the bouncing back property of the material back to its baseline position after its temporarily deformed. This is already considered strictly with respect to the mechanical working of the elastic material.

Repeatability: It is stated as the consistency of repeating same value under same conditions. Since this precision property ties itself with the how many times the actual reading can repeat under same conditions, this can also be considered just mechanically how much volume of water as a whole does an elastic material hold in certain conditions and not the repeatability of an actual reading or a measured true pressure value.

That is, a result is not inaccurate when Linearity, Hysteresis, or Repeatability of a pressure gauge is affected but a strict pointing towards mechanical wear and tear (including effects of temperature, if any).

Design a Pressure Gauge to be mounted vertically (perpendicularly) to avoid flow velocity effects to ensure accurate reading:

ASME design standard B40.100 has following standards for physical design and installation of the pressure gauge:

1. To have the pressure gauge inlet fused with the pressure vessel in such a way it does not protrude (come off) by few inches from the internal area of the pressure vessel.
2. To have pressure gauge installed perpendicularly to measure accurate “Static Pressure” and avoid the velocity effects of the liquid flow.

Consider an example of a water pipeline with a gauge attached as shown in the figure below,

The water that enters the inlet of the gauge is the same water that is flowing in a direction parallel to the pipeline as in, in the forward direction. The only difference is the amount of water. If the gauge were placed directly in the direction of the flow of water the amount of water entering the gauge’s internal elastic material would be actually constant and more. Meaning the gauge in the direction of the flow of water would constantly show the higher or highest reading in the scale.

When switched to the vertical mounting position, the amount of water that enters the elastic material of the gauge is now less as compared to when it was placed in the horizontal direction. The only difference is the lesser volume of water that enters the inlet of the gauge and thus on/ into the elastic material. But lesser volume of water entering the gauge does not mean it is giving an accurate or an actual reading as much as horizontal placement of a gauge does not give an accurate or an actual reading. And as also seen in the [Issue 2] that there is no self-contained “Static or Dynamic Pressure” and how individual containment of water impacts the surface that is containing the water. So measuring the “pressure” perpendicularly(vertical to the flow) is no different than measuring the “pressure” horizontally.

Vertical Position of Gauge ≠ Accurate Reading or (Measure of “Static Pressure”).

Related Issues:

[Issue 1] - Pressure Gauges measure Static Pressure and Gauge Pressure.
[Issue 2] - Pressure in fluids (liquids) = (Static Pressure, Dynamic Pressure).
[Issue 3] - Hydrostatic Pressure = ρgh.
[Issue 4] - Absolute Pressure = Gauge Pressure + Atmospheric Pressure.

Referenced Resources:

"https://generalinstruments.co.in/blog/understanding-pressure-gauge-accuracy-and-calibration"

"https://www.instrumentationblog.in/pressure-gauge-accuracy-classes"

"https://www.instrumart.com/pages/539/pressure-gauge-accuracy-grades"

"https://www.aimscontrols.com/pressure-gauge-accuracy-class-selection-guide"

"https://www.inspection-for-industry.com/maximum-allowable-working-pressure.html"

"https://whatispiping.com/determination-of-design-pressure-design-temperature"

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"https://bcstgroup.com/understanding-pressure-gauge-range"

"https://industrialmonitordirect.com/blogs/knowledgebase/pressure-vessel-design-vs-operating-vs-mawp-terminology"

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